

9(a)	1 NAD ; 2 <u>coenzyme A</u> ;	2
9(b)	any three from: 1 enzyme activity decreases over time at both concentrations (of hydrogen peroxide) ; 2 activity decreases, more / faster / at a greater rate, at high(er) H₂O₂ concentration or enzyme less active with, with high(er) H₂O₂ concentration ; 3 paired data quote ; 4 H₂O₂, acts as an inhibitor / inhibits enzyme activity ;	3

Question	Answer	Marks
9(c)	any four from: 1 energy is used to <u>pump</u>, protons / H^+, into intermembrane space ; 2 proton gradient (formed) ; 3 protons <u>diffuse</u> through ATP synthase ; 4 ATP produced from ADP and P_i ; 5 oxygen accepts electrons to form water ; 6 <i>ref. to chemiosmosis</i> ;	4
9(d)	any three from: 1 respiration in anaerobic conditions ; 2 produces lactate ; 3 need oxygen, to oxidise lactic acid to CO_2 and H_2O / for breakdown of lactic acid in Krebs cycle plus oxidative phosphorylation / (for ETC) to make extra ATP ; 4 ATP used for lactate $\rightarrow$ glucose / glycogen ; 5 <i>idea of re-paying oxygen debt</i> ;	3